ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) path to success

DISTANCE LEARNING PROGRAMME

(Academic Session : 2024 - 2025)

JEE (Advanced)
AIOT
04-05-2025

JEE(Main + Advanced) : LEADER TEST SERIES / JOINT PACKAGE COURSE

Test Type : Full Syllabus

ANSWER KEY — PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	A	B	D	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	C,D	B,C	A,B,C	A,B,C	A,B,D	B,C,D		
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	20	400	4	7	58	738	1	306

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4				
	A.	B	C	C	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,B,C	A,B,C	A,B,D	C,D	A,B,D	A,B,D		
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	6	5	101	8	4	2	5	5

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	D	B	C	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,B,C,D	B,D	A,C	A,C	A,B,C	A,B,C		
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	8	1	3	6	8	9	3	9

HINT – SHEET

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PART-1 : PHYSICS

SECTION-I (i)

2. Ans (B)

$$\theta = \frac{1.22\lambda}{D} = \frac{s}{h}$$

$$s = 2.24 \text{ m}$$

3. Ans (D)

The slope of the φ-x graph is equal to the negative value of the electric field intensity (−E), with the direction of the field being opposite to the slope direction. The field created by a positive charge has a positive potential, while the field created by a negative charge has a negative potential. If multiple field sources exist, the total potential is the algebraic sum of their potentials.

The slope of the φ-x graph is equal to the electric field intensity value −E. Statement C is incorrect because the slope at point N is not zero, meaning the electric field at point N is not zero, so statement A is incorrect. The slope of the tangent at point C is zero, indicating the field at C is zero. Since A and N are located to the right of point M, it can be deduced that $q_1 > q_2$, making statement B incorrect.

From point N to point D, the potential first rises and then falls. Thus, if a negative charge is moved from N to D, the electric force will first do positive work and then negative work, making statement D correct.

PART-1 : PHYSICS

SECTION-II

4. Ans (7)

We note the points at which the curve is zero ($\theta_2 = 60°$ and $140°$). We infer that sheet-2 is perpendicular to one of the other sheets at $\theta_2 = 60°$ and that it is perpendicular to the other of the other sheets when $\theta_2 = 140°$. Without loss of generality, we choose $\theta_1 = 150°$, $\theta_3 = 50°$. Now, when $\theta_2 = 90°$, it will be $|\Delta\theta| = 60°$ relative to sheet-1 and $|\Delta\theta'| = 40°$ relative to sheet-3. Therefore,

$$\frac{I_f}{I_i} = \frac{1}{2}\cos^2(\Delta\theta)\cos^2(\Delta\theta') = 7\%$$

PART-2 : CHEMISTRY

SECTION-I (i)

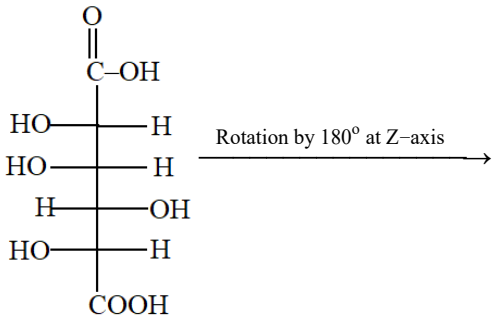
2. Ans (C)

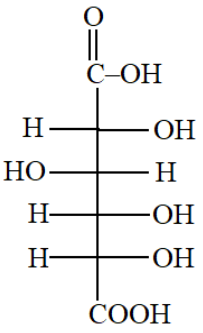
$$K_{A_1} = K_1 : K_{A_2} = 2K_2 : K_{A_3} = 3K_3$$

$$T_{avg.} = \frac{1}{K_A} = \frac{1}{K_1 + 2K_2 + 3K_3} = 8$$

3. Ans (C)

Oxidation product of C is

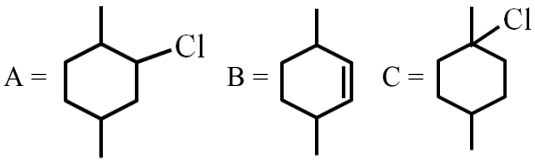




It is also oxidation product of D-glucose

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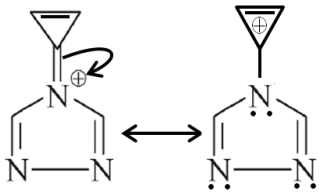
4. Ans (B)



PART-2 : CHEMISTRY

SECTION-I (ii)

9. Ans (A,B,D)



Both rings are aromatic

∴ minimum rotational barrier energy and due to easy breaking of π-bond it is most basic

* All compounds are bicyclic (Not bicyclo)

PART-2 : CHEMISTRY

SECTION-II

4. Ans (8)

Energy of n photons = $nh\upsilon_0$

All this energy converts to KE of Xeton when Xteon absorbs these photons

$KE = nh\upsilon_0 = \frac{1}{2}m_{zeton}V^2$

$nh\upsilon_0 = \frac{1}{2}(4m)V^2$

Hence $v = \sqrt{\frac{nh\upsilon_0}{2m}}$

$\lambda_{debroglie} = \frac{h}{(4m)v} = \frac{h}{4m\sqrt{\frac{nh\upsilon_0}{2m}}} = \frac{\sqrt{h}}{8\sqrt{m\upsilon_0}}$

$\Rightarrow \frac{\sqrt{h}}{4\sqrt{\frac{n}{2}}\sqrt{mv_0}} = \frac{1}{8}\sqrt{\frac{h}{mv_0}} \Rightarrow \frac{1}{8} = \frac{1}{4\sqrt{\frac{n}{2}}}$

$\Rightarrow 2\sqrt{\frac{n}{2}} \Rightarrow n = 8$

5. Ans (4)

$T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$

$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$

$T_2 = 300\,(8)^{2/3} = 1200\ K$

$\frac{T_3}{T_2} = \frac{P_3}{P_2} = \frac{4}{3}$, $T_3 = 1600\ K$

$\frac{T_4}{T_3} = \left(\frac{V_2}{V_1}\right)^{\gamma-1} = \frac{1600}{4} = 400\ K$

$|X| = \left|\frac{W_{234}}{W_{412}}\right| = \left|\frac{W_{34}}{W_{12}}\right| = \left|\frac{\Delta U_{34}}{\Delta U_{12}}\right| = \left|\frac{T_4 - T_3}{T_2 - T_1}\right|$

$|X| = \frac{1200}{900} = \frac{4}{3}$

$3|X| = 4$

6. Ans (2)

$n_{N_2} = \frac{1.135}{22.7} = 0.05\ mol = n_{N_2H_4}$ (n.f. = 4)

eq. N_2H_4 = eq. of I_2 = eq. of PbO_2

0.1 mole of PbO_2

$PbO_2 + 2NaOH \longrightarrow Na_2PbO_3 + H_2O$

0.1 mole $\frac{1}{10}M$

$V = ?$

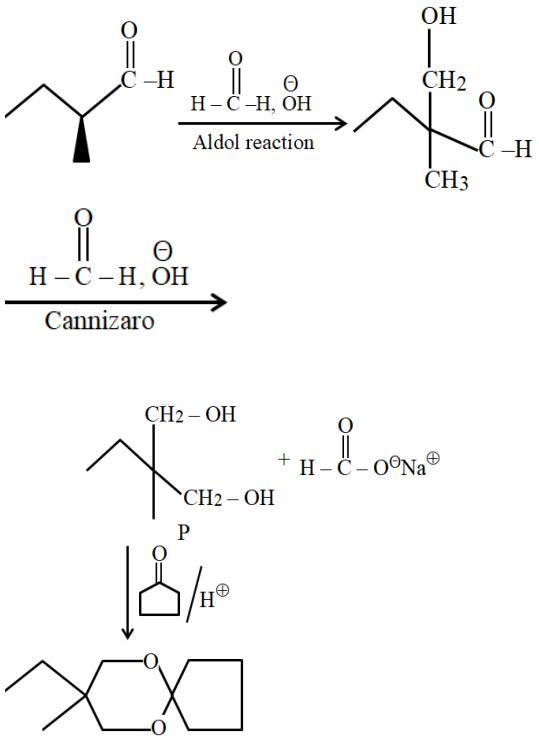
0.1 mol $PbO_2 \longrightarrow$ 0.2 mole NaOH

$0.2 = \frac{1}{10} \times V$

V = 2 litre.

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7. Ans (5)



Value of X = 7

Value of Y = 2

∴ X – Y

= 7 – 2 = 5

8. Ans (5)

i, ii, iv, vii, viii

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (D)

$f'(x) = 3x^2 + 2x + 3 + 2\cos x$

$= \underbrace{3\left(x+\frac{1}{3}\right)^2}_{\geq 0} - \underbrace{\frac{1}{3} + 3 + 2\cos x}_{> 0}$

f (x) is strictly increasing and f (0) = 0

2. Ans (B)

$y = \ell n^2 x - 1$

$y' = \frac{2\ln x}{x} = 0 \Rightarrow x = 1$

$x > 1, y \uparrow$ and $0 < x < 1, y$ is $\downarrow$

$A = \left|\int_{1/e}^{e} (\ln^2 x - 1)dx\right|$

$= \left|x\ln^2 x\right]_{1/e}^{e} - 2\int_{1/e}^{e}\left(\frac{\ln x}{x}\right)\cdot x\,dx - \left(e - \frac{1}{e}\right)\right|$

$= \left|\left(e - \frac{1}{e}\right) - 2\int_{1/e}^{e}\left(\frac{\ln x}{x}\right)\cdot x\,dx - \left(e - \frac{1}{e}\right)\right|$

$= \left|-2\left[x\ln x\right]_{1/e}^{e} - \int_{1/e}^{e} dx\right] = \left|-2\left[\left(e + \frac{1}{e}\right) - \left(e - \frac{1}{e}\right)\right]\right|$

$= \left|\frac{4}{e}\right| = \frac{4}{e}$

3. Ans (C)

$(1 + x + x^3)^{10} = {}^{10}C_0 + {}^{10}C_1 (x + x^3) +$

${}^{10}C_2 x^2 (1 + x^2)^2 + {}^{10}C_3 x^3 (1 + x^2)^3 + \ldots\ldots$

$\ldots\ldots {}^{10}C^9 x^9 (1 + x^2)^9 + {}^{10}C_{10} x^{10} (1 + x^2)^{10}$

$= a_0 + a_1 x + a_2 x^2 + \ldots\ldots + a_{28} x^{28} + \underbrace{a_{29} x^{29}}_{\times} +$

$a_{30} x^{30}$

The expansion does not contain any term of x^{29}

∴ Number of different terms is 30

4. Ans (B)

$(1 + x + 2x^2 + 4x^4)20$

$T_{r+1} = {}^{20}C_r \cdot (1 + x + 2x^2)^{20-r} \cdot (4x^4)^r$

For r = 19

$T_{19+1} = {}^{20}C_{19} \cdot (1 + x + 2x^2)^1 \cdot (4x^4)^{19}$

∴ Coefficient of $x^{78} = 2 \cdot {}^{20}C_{19} \cdot 4^{19} = 2 \cdot 20 \cdot 2^{38} = 10 \cdot 2^{40}$

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PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (A,B,C,D)

$z_1\bar{z}_2 = z_2\bar{z}_3 = z_3\bar{z}_1$ ....(2)

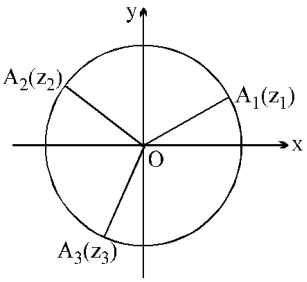
$\therefore \quad |z_1||z_2| = |z_2||z_3| = |z_3||z_1|$

$\therefore \quad \frac{1}{|z_3|} = \frac{1}{|z_1|} = \frac{1}{|z_2|}$

$\therefore \quad |z_1| = |z_2| = |z_3| = r$

$\Rightarrow$ z_1, z_2, z_3 lie on a circle of radius r.

$|z_1|^2 = r^2 \quad \Rightarrow \bar{z}_1 = \frac{r^2}{z_1}$; |||ly $\bar{z}_2 = \frac{r^2}{z_2}$; $\bar{z}_3 = \frac{r^2}{z_3}$



from (2) $z_1 \frac{r^2}{z_2} = z_2 \frac{r^2}{z_3} = z_3 \frac{r^2}{z_1}$

$\frac{z_1}{z_2} = \frac{z_2}{z_3} = \frac{z_3}{z_1}$

hence amp z_1 – amp z_2 = amp z_2 – amp z_3

= amp z_3 – amp z_1

$\Rightarrow$ $A_1A_2A_3$ is equilateral triangle.

$\therefore$ (2) $\Rightarrow$ (1) $\Rightarrow$ (A)

Also $\frac{z_1}{z_2} = \frac{z_2}{z_3} = \frac{z_3}{z_1}$

hence $z_2^2 = z_3z_1$ and $z_3^2 = z_1z_2$ & $z_1^2 = z_2z_3$

$\therefore$ (2) $\Rightarrow$ (3) $\Rightarrow$ (C)

6. Ans (B,D)

We have

$2[x] + 64 = 3[x] - 192 \quad \Rightarrow \quad [x] = 256$

$\Rightarrow$ $x \in [256, 257)$

Also, $\beta = \sin\frac{\pi}{18} \cdot \sin\frac{3\pi}{18} \ldots \sin\frac{9\pi}{18} \ldots \sin\frac{17\pi}{18}$

$= \sin^2\frac{\pi}{18} \cdot \sin^2\frac{3\pi}{18} \cdot \sin^2\frac{5\pi}{18} \cdot \sin^2\frac{7\pi}{18} = \sin^2 10^\circ \cdot \sin^2 30^\circ \cdot \sin^2 50^\circ \cdot \sin^2 70^\circ$

$\frac{1}{4}(\sin 10^\circ \cdot \sin 50^\circ \cdot \sin 70)^2 = \frac{1}{4}\left(\frac{\sin 30}{4}\right)^2$

$\therefore \beta = \frac{1}{4}\left(\frac{1}{8}\right)^2 = \frac{1}{256}$

7. Ans (A,C)

(A) $5\sec^{-1}\alpha + 10\sin^{-1}\beta = 10\pi \quad \sec^{-1}\alpha = \pi$ &

$\sin^{-1}\beta = \pi/2$

$\Rightarrow \alpha = -1; \ \beta = 1$ So $\tan^{-1}(-1) + \cos^{-1}(0)$

$= -\frac{\pi}{4} + \frac{\pi}{2} = \frac{\pi}{4}$

(B) $\sin 2\pi x + \cos 2\pi x = \sin 2\pi x + \frac{1}{2} + \frac{\cos 2\pi x}{2}$

Min. value = $\frac{-\left(\sqrt{5}+1\right)}{2}$

(C) We have $x = \frac{\sqrt{3}\cos 20^\circ - 4\sin 20^\circ.\cos 20^\circ}{\sin 20^\circ}$

$= \frac{2\sin 60^\circ.\cos 20^\circ - 2\sin 40^\circ}{\sin 20^\circ}$

$= \frac{2\sin 60^\circ.\cos 20^\circ - 2\sin 40^\circ}{\sin 20^\circ} = 1$

(D) $f(x) = \sum_{n=1}^{4}\left[1 + \sin\frac{x}{n}\right] = \sum_{n=1}^{4}\left(1 + \left[\sin\frac{x}{n}\right]\right)$

$= 4 + [\sin x] + \left[\sin\frac{x}{2}\right] + \left[\sin\frac{x}{3}\right] + \left[\sin\frac{x}{4}\right]$

$\left[\sin\frac{x}{3}\right]$ and $\left[\sin\frac{x}{4}\right]$ is zero in $(0, p)$

Hence $f(x) = 4 + [\sin x] + \left[\sin\frac{x}{2}\right]$

Now range of $f(x)$ is $\{4, 5\}$

Hence sum is $4 + 5 = 9 \Rightarrow$ (D) is incorrect.

8. Ans (A,C)

(A) $f(x) = \sin^{-1}\left(\log_3\left(\frac{x^2-x+1}{x^2+x+1}\right)\right)$

$f(-x) = \sin^{-1}\left(\log_3\left(\frac{x^2+x+1}{x^2-x+1}\right)\right) = -f(x)$

$f\left(\frac{1}{x}\right) = f(x) = -f(-x)$

\+ – +

–1 1

max. at $x = -1$ min at $x = 1$

(B) $f'(x) = \frac{1}{\sqrt{1-\left(\log_3\left(\frac{x^2-x+1}{x^2+x+1}\right)\right)^2}} \cdot \frac{\log_3 e}{\left(\frac{x^2-x+1}{x^2+x+1}\right)} \cdot \frac{2(x^2-1)}{(x^2+x+1)^2}$

Since $\frac{x^2-x+1}{x^2+x+1} \in \left[\frac{1}{3}, 3\right]$

$\therefore f(x) = \left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$

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9. **Ans (A,B,C)**

Given, $A + adj(A) = \alpha A^T$ .....(1)

(A) If $\alpha = 1$, $A + adj(A) = A^T$

$\Rightarrow adj(A)$

$= -(A - A^T)$

$\Rightarrow |adj(A)| = |-(A - A^T)| = 0$

$\because A - A^T$ is skew symmetric.

(B) If $\alpha = 1$, $A + adj(A) = A^T$

$\Rightarrow 3A + adj(A) = A^T + 2A$

So, $tr(3A + adj(A)) = tr(A^T + 2A) \equiv tr(3A)$

$= 9$ (diagonal element are same in A^T and A).

(C) If $\alpha = -2$, $A + adj(A) = -2A^T$

$\Rightarrow 3A + adj(A) = -2A^T + 2A = 2(A - A^T)$

$\Rightarrow |3A + adj(A)| = 0$

(D) If $\alpha = -2$, $A + adj(A) = -2A^T$

$\Rightarrow 4A + adj(A) = 3A - 2A^T$

So, $tr(4A + adj(A)) = tr(3A - 2A^T) \equiv tr(A) = 3$.

10. **Ans (A,B,C)**

(A) 3^5 ; There are 5 pair of numbers having sum equals to 11

(1, 10); (2, 9); (3, 8); (4, 7); (5, 6)

from every pair, one number can be taken in 3 ways

Hence number of subsets $3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 = 35$

(B) 3^5 ; $xyz = 2310 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11$

2 can be distributed to x, y, z in 3 ways.

3 can be distributed to x, y, z in 3 ways.

5 can be distributed to x, y, z in 3 ways.

7 can be distributed to x, y, z in 3 ways.

11 can be distributed to x, y, z in 3 ways.

Hence total ways is 3^5

(C) 3^5;

1					

1st place can be filled only in one way i.e. 1

remaining 5 places can be filled in 3^5 ways

Hence total number of ways $= 1 \cdot 3^5 = 3^5$

Ans.

(D) $3^5 - 1$; Obvious.

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (8)**

Equation of normal at $P_1(4\cos\theta_1, 3\sin\theta_1)$ is

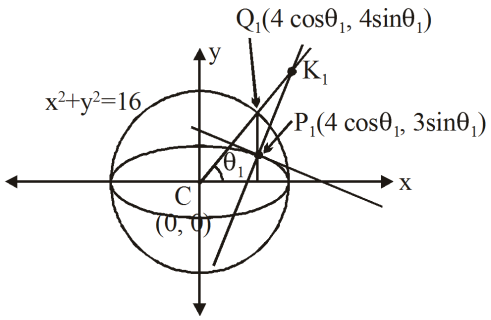
$$\frac{4x}{\cos\theta_1} - \frac{3y}{\sin\theta_1} = 7 \qquad \ldots (1)$$

Also, equation of CQ_1 is

$$y = \left(\frac{\sin\theta_1}{\cos\theta_1}\right)x \qquad \ldots (2)$$

$\therefore$ Solving (1) and (2), we get

$$\frac{4x}{\cos\theta_1} - \frac{3}{\sin\theta_1}\left(\frac{\sin\theta_1}{\cos\theta_1}\right)x = 7$$



$$\Rightarrow \frac{x}{\cos\theta_1} = 7 \quad \Rightarrow \quad x = 7\cos\theta_1,\ y = 7\sin\theta_1$$

So, $K_1 = (7\cos\theta_1, 7\sin\theta_1) \Rightarrow CK_1 = 7$

Similarly, $CK_2 = CK_3 = ... = CK_n = 7$

$$\therefore \sum_{i=1}^{n} CK_i = 56 \Rightarrow 7n = 56 \Rightarrow n = \frac{56}{7} = 8$$

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2. Ans (1)

$$\frac{f(x)}{1+x^2} = 1 + \int_0^x \frac{f^2(t)\,dt}{1+t^2} \qquad [\text{Note: } f(0) = 1]$$

differentiate both sides w.r.t. x

$$\frac{(1+x^2)f'(x) - f(x)(2x)}{(1+x^2)^2} = \frac{f^2(x)}{(1+x^2)}$$

$$\frac{dy}{dx} - \left(\frac{2x}{1+x^2}\right) y = y^2$$

$$\frac{1}{y^2}\frac{dy}{dx} - \left(\frac{2x}{1+x^2}\right)\frac{1}{y} = 1$$

Let $\frac{-1}{y} = t$

$$\frac{dt}{dx} + \left(\frac{2x}{1+x^2}\right) t = 1$$

solving the above L.D.E,

$$f(x) = \frac{-3(1+x^2)}{x^3+3x-3}$$

$$f(-2) = \frac{-3(5)}{-8-6-3} = \frac{15}{17}$$

3. Ans (3)

$$I = \int_{\frac{-1}{\sqrt[3]{n}}}^{\frac{1}{\sqrt[3]{n}}} x\,\ell n\,(1+e^x+e^{2x}+\ldots\ldots\ldots e^{2nx})\,dx \quad \ldots(1)$$

Using k ing

$$I = \int_{\frac{-1}{\sqrt[3]{n}}}^{\frac{1}{\sqrt[3]{n}}} -x\,\ln\left(1+\frac{1}{e^x}+\frac{1}{e^{2x}}+\ldots\ldots\ldots+\frac{1}{e^{2nx}}\right) dx$$

$$= \int_{\frac{-1}{\sqrt[3]{n}}}^{+\frac{1}{\sqrt[3]{n}}} -x\,\ln\left(\frac{e^{2nx}+e^{(2n-1)x}+\ldots\ldots\ldots+e^x+1}{e^{2nx}}\right) dx$$

$$I = -I + \int_{\frac{-1}{\sqrt[3]{n}}}^{+\frac{1}{\sqrt[3]{n}}} x\,\ln e^{2nx}\,dx$$

$$\therefore 2I = 2n\left(\frac{x^3}{3}\right)_{\frac{-1}{\sqrt[3]{n}}}^{\frac{1}{\sqrt[3]{n}}} = 2n\left(\frac{1}{3n}+\frac{1}{3n}\right) = \frac{4}{3}$$

$$\therefore I = \frac{2}{3}$$

4. Ans (6)

W_1: $C_1 = (-5, 12)$ W_2: $C_2 = (5, 12)$

$r_1 = 16$ $r_2 = 4$

now, $CC_2 = r + 4$

$CC_1 = 16 - r$

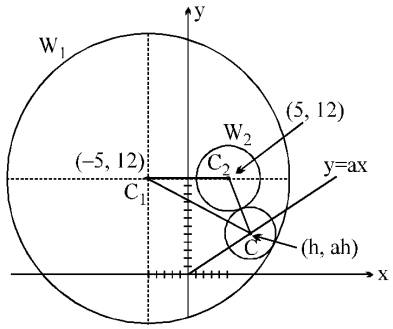
let C(h, k) = c(h, ah)

$CC_1^2 = (16-r)^2$

$\Rightarrow (h+5)^2 + (12-ah)^2 = (16-r)^2$

$CC_2^2 = (4+r)^2$

$\Rightarrow (h-5)^2 + (12-ah)^2 = (4+r)^2$



By subtraction

$20h = 240 - 40r$

$\Rightarrow h = 12 - 2r \Rightarrow 12r = 72 - 6h \quad \ldots(1)$

By addition

$2[h^2 + 25 + a^2h^2 - 24ah + 144]$

$= 272 - 24r + 2r^2$

$h^2(1+a^2) - 24ah + 169 = 136 - 12r + r^2$

$= 136 + (6h - 72) + \left(\frac{12-h}{2}\right)^2$ [using (1)]

$\Rightarrow 4[h^2(1+a^2) - 24ah + 169] = 4[64 + 6h] +$

$(12-h)^2 = 256 + 144 + h^2$

$\Rightarrow h^2(3+4a^2) - 96ah + 105 \cdot 4 - 36 \cdot 4 = 0$

$\Rightarrow h^2(3+4a2) - 96ah + 69 \cdot 4 = 0;$

for 'h' to be real $D \geq 0$

$\Rightarrow (96a)^2 - 4 \cdot 4 \cdot 69\,(3+4a^2) \geq 0$

$\Rightarrow 576a2 - 69.3 - 276a^2 \geq 0$

$300a^2 \geq 207 \Rightarrow a^2 \geq \frac{69}{100}$;

hence m (smallest) = $\frac{13}{10}$

So, $m^2 = \frac{69}{100}$; $\therefore$ p + q = 169

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5. Ans (8)

$$f\left(0^-\right) = \lim_{x\to 0} \frac{x\left[\frac{\ell n(1+x)}{x} - \frac{\ell n(1-x)}{-x}\right]x}{x^2 \frac{(1-\cos x)}{x^2}\left(\frac{1+\cos x}{\cos x}\right)} = 0$$

& $f(0^+) = 0$

$\therefore$ function is always continuous.

$$f\left(0^-\right) = \lim_{h\to 0} \frac{f(0-h)-f(0)}{-h}$$

$$= \lim_{h\to 0} -h\left(\frac{\ell n(1-h)+\ell n(1+h)}{(1-\cos^2 h).\frac{1}{\cosh}}\right) = 0$$

& $f'(0^+) = k^2 - 3k - 1$

$\because$ function are differentiable at x = 0

$\Rightarrow k^2 - 3k - 1 = 0 \begin{cases} k_1 \\ k_2 \end{cases}$

$k_1 + k_2 = 3,\ k_1k_2 = -1$

$\therefore k_1^2 + k_2^2 = (k_1 + k_2)^2 - 2k_1k_2 = 11$

6. Ans (9)

$\log_3 |x-2| = 2^{\log_2\left(\log_3(|x|-2)\right)}$

$\Rightarrow \log_3 |x-2| = \log_3(|x|-2)$

$\Rightarrow |x-2| = |x| - 2$

$\Rightarrow |x| - |x-2| = 2$

Case-I : $x < 0$

$-x + x - 2 = 0$

$-2 = 2$ (no solution)

Case-II: $0 \le x < 2$

$x + (x-2) = 2$

$\Rightarrow 2x = 4 \Rightarrow x = 2$ (no solution)

Case-III : $x \ge 2$

$x - (x-2) = 2$

$\Rightarrow 2 = 2$

$\therefore x \ge 2$

But $\log_3(|x|-2) > 0 \Rightarrow |x| - 2 > 1 \Rightarrow |x| > 3$

$\therefore x \in (3, \infty)$ is the solution set of the equation

Least integral value of x is 4.

Now, $(\alpha)^{-4\alpha} = (4)^{-16} = 2^{-32}$

$N = 2^{-32}$

$\log_{10} N = -32 \log_{10} 2 = -32 \times 0.301 = -9.632$

$\therefore$ Number of zeros after decimal and before first significant digit in 2^{-32} is 9

7. Ans (3)

det. (A) $= x_1 \cdot x_2 \cdot x_3$

det. (B) $= y_1 \cdot y_2 \cdot y_3$

$\therefore x_1 \cdot x_2 \cdot x_3 = y_1 \cdot y_2 \cdot y_3$

Both sides	Number of solutions.	
0	$({}^3C_1 \cdot 2 \cdot 2 + {}^3C_2 \cdot 2 + {}^3C_3)^2 = 19^2$	$= 361$
1	$({}^3C_2 \cdot 1 + {}^3C_3)^2 = 4^2$	$= 16$
-1	$({}^3C_2 \cdot 1 + {}^3C_3)^2 = 4^2$	$= 16$
		$N = 393$

$\therefore \frac{N}{131} = 3$. Ans.]

8. Ans (9)

$y^{2/3} - 2xy^{1/3} + 1 = 0$

$y^{1/3} = \left(x \pm \sqrt{x^2-1}\right) \Rightarrow \ell n\ y = 3\ \ell n \left(x \pm \sqrt{x^2-1}\right)$

$\frac{y_1}{y} = \frac{\pm 3}{\sqrt{x^2-1}} \Rightarrow (x^2-1)\,y_1^2 = 9y^2$

$2x\,y_1^2 + (x^2-1)\,2y_1y_2 = 18yy_1$

$xy_1 + (x^2-1)y_2 = 9y$ (As y_1 is not equal to 0, because y is not constant)

Dividing by y, we get

$\therefore\ x\frac{y_1}{y} + (x^2-1)\frac{y_2}{y} = 9$ Ans.

Alternatively: Given $2x = \left(y^{\frac{1}{3}} + y^{\frac{-1}{3}}\right)$ .....(1)

Differentiate both side with respect to x, we get

$$2 = \left(\frac{1}{3}y^{\frac{1}{3}-1} - \frac{1}{3}y^{\frac{-1}{3}-1}\right)y_1$$

$$\Rightarrow 2 = \frac{\left(y^{\frac{1}{3}} - y^{\frac{-1}{3}}\right)y_1}{3y}$$

$\Rightarrow 6y = \left(y^{\frac{1}{3}} - y^{\frac{-1}{3}}\right)y_1$ (squaring)

$\Rightarrow 36y^2 = (4x^2-4)y_1^2$ {using equation (1) }

$\Rightarrow 9y^2 = (x^2-1)y_1^2$ .....(2)

Again differentiate both sides of above equation, we get

$18yy_1 = (x^2-1)\cdot 2\,y_1y_2 + 2xy_1^2$

$\Rightarrow 9y = (x^2-1)y_2 + xy_1$ (As y is not constant so $y_1 \ne 0$)

$$\Rightarrow 9 = \frac{(x^2-1)}{y}\cdot y_2 + \frac{x}{y}\cdot y_1$$

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